ALSON LEE JIAN YANG

alsonleej@u.nus.edu • +65 86864036 • GitHub • LinkedIn • alson-portfolio.vercel.app • Singapore

TECHNICAL SKILLS

Programming Languages | Java, TypeScript, JavaScript, HTML, CSS, SQL, MATLAB, Microsoft VBA, R, Python, C, C++
Frameworks | React, TailwindCSS, Bootstrap, Phaser.js, tRPC, Next.js, Flask, Bottle
Developer Tools | Vite, Next.js, Jest, JUnit, Linux/Unix, Shell, Zod, Drizzle, Axios, Docker
Cloud & DevOps | Git/GitHub, Vercel, AWS (Amplify, Lambda, Cognito, DynamoDB, S3, API Gateway), GCP, Terraform
Visual Design | Canva, Fusion360, Tiled, Google Stitch

EXPERIENCE

Fullstack Software Engineer Intern, Kisi – Stockholm, Sweden (NUS Overseas Colleges) — Jan 2026 — Present

Security technology company leading access control solutions — TypeScript | React | TailwindCSS | GCP | Docker | Terraform

- Built a live **leaderboard and analytics** app on a **Raspberry Pi**, connected to a Concept2 SkiErg and Kisi readers, presented at **FIBO Conference** in Cologne to **367** participants
- Migrated a third-party camera integration from **Firestore** to **PostgreSQL**, with **no downtime**, using dual-writes, a backfill and staged read cutover, saving **$500/mth** on infrastructure costs across **600+** customers
- Built and own an integration with Rhombus, an enterprise security camera company, end to end (React, Node.js, PostgreSQL, Terraform), and **migrated** existing customers onto it
- Owned **4 production apps**: visitor check-in kiosks, room and desk booking, badge printing, and camera integration
- Shipped a **custom form builder** for **2 products** within a week of a customer request, covering the schema, shared backend, admin UI and docs
- Provisioned **GCP secrets** via Terraform for secure deployment

Founding Software Engineer, VisEyeQ – Singapore — Feb 2025 — Dec 2025

VC-backed AI startup revolutionising warehouse safety powered by IoT — TypeScript | React | TailwindCSS | Vite | AWS

- Developed a **B2B SaaS cloud dashboard** analysing data from IP cameras, IoT wearables and ML models
- Managed persistent user/incident data with **AWS DynamoDB**; streamed video from AWS S3
- Engineered **secure authentication** using AWS Cognito with email and Google OAuth
- Integrated IoT wearables with frontend using **custom APIs** via AWS API Gateway
- Built **AWS Lambda** functions to compute injury cost savings and ROI
- Iterated UI/UX with clients to prioritise functionality and intuitive use

Software Engineer, Source Academy – Singapore — Jan 2025 — May 2025

Online programming environment and visual debugger to teach programming classes — TypeScript | React | Jest

- Implemented **7 bugfixes** and **3 new features** used by **1,200+** undergraduates
- Wrote **10 high-coverage Jest** test cases for unit/integration testing
- Co-developed **CSE Machine**, a Source-to-JavaScript interpreter
- Engineered core features: **control flow**, **instruction order**, and **built-ins**
- Selected presenter at **NUS STePS conference**

Signal Operator, Singapore Armed Forces (Lancer) – Brunei Darussalam — Jan 2022 — May 2024

- Grew broadband coverage **200%** and **3×** capacity by upgrading camp networks to WiFi 6
- Installed and maintained industrial networking hardware and software under operational constraints
- Cut admin form time **50%** and eliminated recurring human errors by building an Excel VBA macro

TECHNICAL PROJECTS

Jobber — 2026

Agentic orchestrator for automated, tailored job applications — Claude | Cursor | MCP

- Engineered an **agentic workflow** that sources postings, tailors a resume per role, and submits applications end-to-end
- Wrote custom **Skills** to generate per-application resumes by matching job descriptions against a structured record of past work
- Ingested listings from **Greenhouse**, **Ashby**, and **LinkedIn** via a unified scraper interface

- Cut time from posting to submitted application to **<24h**, across **50+** applications
- Orchestrated a **browser sub-agent** and **PDF generation** pipeline over **MCPs**

Shift Happens **2025**

Web puzzle-platformer game JavaScript | Phaser.js | Tiled | Vite | Vercel

- Built an **8-level** puzzle-platformer in Phaser.js with fluid movement and powerups
- Created a drag-and-drop level editor by wiring **Tiled** JSON maps into Phaser
- Added a generated asset loader to speed up sprite/item preload during development
- Stabilized stacked-block physics with custom collision separation and gravity rules
- Deployed on **Vercel** for browser play without installs

FirstImpressions **Aug 2025 — Nov 2025**

Team & Client Booking Manager Java | JavaFX | JUnit | Gradle | GitHub Actions | Codecov

- Co-developed a **Java** CLI/GUI app to match professionals to clients by skill tags and manage bookings
- Implemented **booking conflict detection** to prevent double-booking
- Added chronological sorting of upcoming vs past appointments
- Set up CI (GitHub Actions, Codecov, checkstyle) across a **4-person** team

EDUCATION

National University of Singapore (NUS), BComp, Computer Science (Hons) **Aug 2024 — Present**

- Year 3 | Expected Graduation: May 2028 | NUS Merit Scholar (Top <5.0%) | GPA: 4.7 / 5 (First Class Honours)
- Internship availability: full time: May - Aug 2027, part time: whole of 2027

Hwa Chong Institution (HCI), GCE 'A' Levels **Jan 2020 — Dec 2021**

- AAA/AAA in Physics, Chemistry, Mathematics, Economics, General Paper, Project Work | H3 Chemistry Merit
- Science & Math Talent Program | Hwa Chong Diploma with Distinction | MOE Edusave Scholarship for Independent Schools (ESIS) Scholar

ADDITIONAL INFORMATION

Languages | English, Mandarin (both spoken and written), Japanese (JLPT N4)
Interests | Entrepreneurship, Reading, Solving puzzles, Hunting for good food, Games, Finance